

Project Bill of Materials (BOM) - 270 Boll Ave

Project Name: 270 Boll Ave

Document Type: Integrated Systems Specification

Integrator Draft: v1.2 (Standardized Infrastructure)

Revision History

Version	Date	Description	Status
v1.0	2026-01-26	Initial Draft (Core Infrastructure)	Archived
v1.1	2026-01-27	Panel Consolidation & Door Entry Re-Spec	Archived
v1.2	2026-01-27	Unified DC Security Island & Environmental Hub	Current

1. Project Objectives & System Organization

The automation system at 270 Boll Ave is designed for maximum reliability, security hardening, and high-fidelity environmental control. The architecture is organized into three localized **Load Control Panels (LCPs)** to minimize home-run copper and isolate failure domains:

- LCP-1 (Garage):** The "Power Hub" for central living areas, kitchen lighting, and high-inrush holiday circuits.
- LCP-2 (Office 2):** The "Environmental Hub" and Master Wing controller. Houses weather station logic, PM2.5 wildfire interlocks, and master bedroom lighting.
- LCP-3 (Tech Room):** The "Security Island." A 100% battery-backed, DC-powered enclosure that manages 3D face recognition and entry without cloud dependency.

Core Standards: All lighting is **DALI-2** (addressable). All logic and switching is **KNX** (distributed). Infrastructure is "Future-Proofed" via stealth pre-wiring for window shades and seasonal exterior lighting.

Section 1: High-Voltage (HV) Infrastructure

Main Service & Smart Panel Telemetry

Part ID	Manufacturer	Description	Qty	Location	Notes
SPAN-GEN2	SPAN	Smart Panel Gen 2 (32-Circuit)	2	Garage	Main Service (Side-by-side)
SPAN-SUB	SPAN	Smart Panel Gen 2 (32-Circuit)	1	North Wall	Sub-Panel for Left Wing
SER-1-AL	Generic	#1 AWG Aluminum SER Cable	-	Field	Sub-SPAN Feed from Main
NM-B-12/2	Southwire	12/2 Romex (High Current Loads)	-	Field	Direct to SPAN (Fridge, Range, etc)

Section 2: Load Control Panel 1 (LCP-1) (Garage) - Kitchen/Main Hub

The automation core for the central living areas.

Part ID	Manufacturer	Description	Qty	MPN	Notes
SAG-24H2408	Saginaw	24x24x8 NEMA 4/12 Enclosure	1	SCE-24H2408LP	Hinged Cover
MW-SDR-480	Mean Well	480W 24V DC PSU (UL 508)	1	SDR-480-24	Main Logic Power
ABB-KNX-PS	ABB	KNX Power Supply 640mA	1	SV/S 30.640.5.1	Powers Line 1.1
ABB-LK-S	ABB	KNX Line Coupler	1	LK/S 4.2	Connects 1.1 to Backbone
PHX-CBM-E4	Phoenix Contact	Electronic Circuit Protector	1	CBM E4 24DC	NEC Class 2 Compliance
SIE-DALI-GW	Siemens	KNX/DALI Gateway Twin	1	N 141/03	Universes 1 & 2
ELD-PWR-100	eldoLED	100W DALI-2 LED Driver	6	1060/A	Garage/Kitchen/Living Tape
LUN-DALI-010	Lunatone	DALI 0-10V PWM Interface	1	86458668	Powder Room Fan Control

Section 3: Load Control Panel 2 (LCP-2) (Office 2) - Master Wing Hub

The automation core for the Master Bedroom, Bedrooms 3/4, and local environmental logic.

Part ID	Manufacturer	Description	Qty	MPN	Notes
HOF-ASE24X12X4	nVent HOFFMAN	Automation Enclosure (24x12x4)	1	ASE24X12X4NK	LCP-2 Top Box (DIN Rail)
HOF-ASE12X12X4	nVent HOFFMAN	Network/Aux Enclosure (12x12x4)	1	ASE12X12X4	LCP-2 Bottom Box (Switches)
ONL-FAC-201	OnLogic	Industrial DIN-Rail PC	1	FR201	Master Logic Controller (LCP-2)
ABB-KNX-PS	ABB	KNX Power Supply 640mA	1	SV/S 30.640.5.1	Powers Line 1.2
ABB-LK-S	ABB	KNX Line Coupler	1	LK/S 4.2	Connects 1.2 to Backbone

SIE-DALI-GW	Siemens	KNX/DALI Gateway Twin	1	N 141/03	Universes 3 & 4
ELD-PWR-100	eldoLED	100W DALI-2 LED Driver	6	1060/A	Master Bed Accent/Coves
LUN-DALI-010	Lunatone	DALI 0-10V PWM Interface	4	86458668	Bath/Laundry Fan Controls

Section 3.1: Load Control Panel 3 (LCP-3) (Tech Room) - Door Entry MDF

The centralized "Security Island" - 100% DC-backed for maximum uptime.

Part ID	Manufacturer	Description	Qty	MPN	Notes
SAG-24H2408	Saginaw	24x24x8 NEMA Enclosure	1	SCE-24H2408LP	Hinged Cover (Security MDF)
ALT-EFLOW104N	Altronix	10A Power Supply/Charger	1	eFlow104N	System Core: Managed DC Rail
BATT-12AH	Yuasa	12V 12Ah SLA Battery	2	NP12-12	24V String (Powers NUC/Switch/Locks)
MW-DDR-60G-19	Mean Well	24V to 19V DC-DC Converter	1	DDR-60G-19	DIN-Rail Power for NUC
ASUS-NUC-13	ASUS	NUC 13 Pro (i5, 16GB RAM)	1	NUC13ANKi5	Security Brain (SDMC Server)
TPL-ISC1008P	TP-Link	Industrial 8-Port PoE+ Switch	1	TL-ISC1008P	DC-Input (12-55V) PoE Source
AKU-SR01	Akuvox	Secure Relay Module	4	SR01	DIN-Rail Mounted via Clips
ABB-BIN-16	ABB	16-fold Binary Input	1	BE/S 4.20.2.1	Security/Status Monitoring

Section 4: KNX Infrastructure (Field)

User Interfaces and Sensors

4.1 Backbone & IP Connectivity

Part ID	Manufacturer	Description	Qty	Notes
ABB-IPS-S	ABB	KNX IP Interface Secure	1	IPS/S 3.1.1 (In LCP-2) - High-Speed Logic Link

ABB-USB-S	ABB	KNX USB Interface	1	USB/S 1.2 (In LCP-2) - Local Diagnostic Backup
ABB-PS-160	ABB	KNX Power Supply 160mA	1	Powers the Main Backbone Line

4.2 Sensors (Environmental & Presence)

Product ID	Manufacturer	Description	Qty	Notes
STE-TP-KNX	Steinel	True Presence Multisensor KNX	4	Primary Bath, Guest, Bath 3, Laundry
ABB-BIN-16	ABB	16-fold Binary Input	2	Door/Window Contacts (In LCPs)

4.2 User Interfaces & Cabling

Part ID	Manufacturer	Description	Qty	Notes
BAS-SENT-4	Basalte	Sentido (4-Button) Switch	12	Minimal code-required placement
BAS-BUS-COU	Basalte	KNX Bus Coupler for Sentido	12	Required for switch connection
CAB-KNX-G	Unitronic	KNX Green Bus Cable (UL)	1	1000ft Spool
WAG-KNX-CONN	Wago	KNX Push-Wire Connectors	2	Red/Black (Boxes of 50)

Section 5: Network & AI Compute (Office 2 / MDF Hub)

Secondary compute and data storage in rack.

Part ID	Manufacturer	Description	Qty	MPN	Notes
ASUS-NUC-AI	ASUS	NUC 13 Pro (i7, 32GB RAM)	1	NUC13ANKi7	AI Compute Node (Pattern Learning/LLM)
SSD-990-4TB	Samsung	990 Pro 4TB NVMe SSD	1	MZ-V9P4T0BW	High-endurance for AI/Logging
UI-SW-PRO-24	Ubiquiti	UniFi Pro Max 24 PoE	1	USW-Pro-Max-24-PoE	Core PoE Backbone
UI-CAM-G5P	Ubiquiti	G5 Professional 4K Camera	12	UVC-G5-Pro	
CYB-PDU-1U	CyberPower	1U Rack PDU	1		Powers Server & Network Gear

APC-UPS-1500	APC	Smart-UPS 1500VA	1	SMT1500C	Backup for Linux Box & AI Nodes
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Section 6: Ventilation & Fans

Whole-House, Attic, and Bathroom Exhaust (EC Motors + DALI-2 Control)

6.1 Whole-House Fans (WHF)

Part ID	Manufacturer	Description	Qty	Notes
QC-TRI-7.0X	QuietCool	Trident Pro 7.0X (EC Motor)	2	1 per arm of U-shape
SUN-2701S-DT7	Sunricher	DALI-2 Relay Puck	2	Power On/Off (In LCP-1/2)
SUN-2401-10V	Sunricher	DALI to 0-10V Signal Converter	2	Variable Speed Control

6.2 Attic Exhaust Fans

Part ID	Manufacturer	Description	Qty	Notes
QC-AFG-3.0	QuietCool	AFG PRO-3.0 Gable Fan	4	2 per gable end
SUN-2701S-DT7	Sunricher	DALI-2 Relay Puck	4	Power On/Off
SUN-2401-10V	Sunricher	DALI to 0-10V Signal Converter	4	Variable Speed Control

6.3 Bathroom & Laundry Exhaust

Location	Fan Motor	Silencer	Backdraft Damper	Diffuser (InviAir)
Primary Bath	Fantech FG 6M EC (Shared)	Fantech LD 6	RSK 6	L100 (60") + S100 (6")
Bath 2 (Kids)	Fantech FG 6M EC	Fantech LD 6	RSK 6	L100 (48")
Bath 3 (Ens)	Fantech FG 6M EC	Fantech LD 6	RSK 6	L100 (48")
Guest Bath	Fantech FG 6M EC	Fantech LD 6	RSK 6	L100 (60")
Laundry Room	Fantech FG 6M EC	Fantech LD 6	RSK 6	S100 (8")
Half Bath	Fantech FG 6M EC	Fantech LD 6	RSK 6	S100 (6")

6.4 Environmental AI Inputs

Part ID	Manufacturer	Description	Qty	Notes
STE-TP-KNX	Steinel	True Presence Multisensor KNX	4	Primary, Guest, Bath 3, Laundry
WAR-WETH-PRO	Warema	KNX Weather Station Pro	1	Outdoor (North Wall)
TEM-PM25	Temco	Outdoor PM2.5 Sensor (Modbus)	1	Wildfire Smoke Interlock
MDT-SCN-MB	MDT	KNX/Modbus Gateway	1	Bridge for PM2.5 Data

Section 7: Field Devices - Lighting (DALI-2)

Based on current floor plan placements.

Product ID	Manufacturer	Description	Qty	Type	Notes
DMF-X2-SQ-FL	DMF Lighting	X-Series Square Flangeless	80	Recessed	Modular: Housing + Trim + LED
DMF-X2-WET	DMF Lighting	X-Series Sq Flangeless (Wet)	4	Recessed	1 per Full Bath (Shower/Tub)
BAF-HAIKU-52	Big Ass Fans	Haiku 52" Aluminum	3	Fan+Light	DALI 0-10V Int (010447)
CAB-NM-PCS	Southwire	Romex PCS Duo (14/2 + 16/2)	3	Spools	Power+Control for Lights & Fans

Section 8: Bulk Infrastructure & Consumables

Item	Manufacturer	Description	Qty	Notes
CAB-KNX-G	Unitronic	KNX Green Bus Cable (UL)	TBD	See Routing Notes Below
CAB-CAT6-RIS	TrueCable	Cat6 Riser (Bulk)	TBD	
CAB-16/4-SHLD	Generic	16/4 Shielded (Shade Pre-wire)	TBD	Home-run to LCP-1/2
BOX-SQ-4IN	Generic	4" Square Steel Box (2-1/8" D)	-	Deep boxes for DALI pucks
BOX-FIRE-PAD	STI	SpecSeal Putty Pad	1	Required for Garage Intercom
CON-EMT-1IN	Generic	1" EMT Conduit	2	Solar Roof-Runs

CON-EMT-2.5IN	Generic	2.5" EMT Conduit	2	Energy Wall Super-Conduit
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Section 9: Door Entry & Access Control (Akuvox)

3D Face Recognition & SIP Intercom System

Part ID	Manufacturer	Description	Qty	MPN	Notes
AKU-X915S	Akuvox	8" 3D Face Recognition Intercom	1	X915S	Front Door (Flush Mount)
AKU-E16C	Akuvox	5" 3D Face Recognition Intercom	3	E16C	Garage Man, Side Door, Laundry Door
AKU-E16-HOOD	Akuvox	Sun Shield / Rain Hood	1	E16-A05H	For Garage Man Door (North Facing)
AKU-S567G	Akuvox	10" Android 12 Indoor Monitor	1	S567G	Hallway Tablet (Main Control)
HES-1006	HES	Heavy Duty Electric Strike	4	1006-CS	24V DC, Fail-Secure
AKU-SR01	Akuvox	Secure Relay Module	4	SR01	(Qty in Section 3.1)

Section 10: Motorized Skylights & HVAC Integration

Centralized 24V DC Actuation & Climate Bridging

Part ID	Manufacturer	Description	Qty	MPN	Notes
MDT-JAL-0810	MDT	8-fold Shutter Actuator (24V DC)	1	JAL-0810.02	Skylights Only
INT-MOD-KNX	Intesis	HVAC Modbus to KNX Gateway	1	INKNXMBM100	Bridge for AC/Heat Recirculation
MW-MDR-100-24	Mean Well	100W 24V DC DIN-Rail PSU	1	MDR-100-24	Dedicated Power for Skylight Motors
WAR-WETH-PRO	Warema	KNX Weather Station Pro	1	#2064965	Home-runs to LCP-2
TEM-PM25	Temco	Outdoor PM2.5 Sensor (Modbus)	1	OAL-PM2.5	Home-runs to LCP-2

Section 11: Routing & Pro-Integrator Notes

11.1 KNX Bus Topology (Redundant Loop Strategy)

- **Perimeter Run:** Route KNX Green Bus through all habitable walls at **48" AFF** (Switch Height).
- **Ceiling/Attic Spine:** Provide a central bus run through the attic with drops to all bathroom sensor locations.
- **Hardened Loop:** All major bus segments must be pulled as a **"Terminated Loop"**. Run the cable from the LCP, through the zone, and back to the LCP. **CRITICAL:** Only one end is to be connected to the bus power supply; the return tail must be labeled and capped.

11.2 LCP Service Spares & Pre-Wire

- **Window Shades (Pre-Wire):** Pull **16/4 Shielded Cable** from LCP-1/2 to every window header. Terminate in "Router & Rescue" vaults per *window-shade-wiring-prep.md*. Coil 3ft of slack in LCP, label, and do not terminate.
 - **LCP-1 (Garage):** Pull **3x Spare Service Loops** (120V HV In + DUO Control Out) into the attic.
 - **LCP-2 (Office 2):** Pull **3x Spare Service Loops** (120V HV In + DUO Control Out) into the attic.
- Environmental Hub:** Weather station and PM2.5 sensors terminate here.

11.3 Box & Finish Standards

- **Deep Boxes:** Standardize on 2-1/8" deep steel boxes to allow for DALI relay pucks and KNX pigtail service loops.
- **Floating Shields:** Drain wires on the KNX bus should be cut and taped at the device end; bond only at the LCP ground rail to prevent ground loops.

11.4 Security Puck (SR01) Placement

- **Anti-Tamper Protocol:** All **Akuvox SR01** security relays must be centralized in **LCP-3** (Tech Room).
- **NO LOCAL LOGIC:** Under no circumstances should the relay be placed near the door terminal. The connection from door to panel is RS485 encrypted data only. This prevents "paperclip" attacks on the lock hardware.
- **Mounting:** All pucks to be DIN-rail mounted using official clips. Label each with the corresponding door name.

Section 12: Patio Infrastructure (Pre-Wire Only)

High-Current Heating & Outdoor Fan Control

Part ID	Manufacturer	Description	Qty	Notes
CAB-10/3-NM	Southwire	10/3 Romex (30A 240V)	3	Home-run from Sub-SPAN to Heater J-Box
CAB-NM-PCS	Southwire	Romex PCS Duo (14/2 + 16/2)	2	Home-run from LCP-2 to Patio Fans
CAB-DALI-G	Generic	18/2 Shielded (DALI Bus)	1	Home-run from LCP-2 to Heater J-Box

12.1 Patio Implementation Notes

- **Heaters (18kW Plan):** Pull 3x 30A dedicated circuits. Control is via DALI-2 relay/contactor logic (hardware deferred).
 - **Fans:** Pull power + control (Duo) for 2x Outdoor Haiku fans.
 - **Surge Protection:** All outdoor data/bus lines must include a DIN-rail surge protector at the LCP entry point.
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Section 13: Known Missing & Future-Phase TBDs

13.1 Aesthetic & Decorative Lighting (Pending Selection)

- **Chandeliers / Pendants:** Dining, Entry, and Kitchen Island fixtures are TBD. **Requirement:** All decorative junction boxes must be 2-1/8" deep to accommodate DALI-2 relay/dimmer pucks.
- **Vanity Sconces:** All bathroom vanity lighting locations and fixtures are TBD.
- **Outdoor Architectural Sconces:** Front entry and perimeter wall sconces are TBD.

13.2 Seasonal & Holiday Lighting (KNX Controlled)

- **Eave Outlets:** 2x Weatherproof outlets mounted under the roof eaves (North/South peaks) for Christmas/Holiday lighting. Home-run to **LCP-1**.
- **Ground Seasonal Outlets:** 2x Weatherproof outlets at ground level (Front/Side) for seasonal displays. Home-run to **LCP-1**.
- **Control Logic:** These 4 outlets must be terminated on a **KNX High-Inrush Relay Actuator** in LCP-1 for automated seasonal scheduling.

13.3 Exterior Infrastructure (Future Wiring)

- **Landscape Spine:** Pull 1" EMT conduit from Garage (LCP-1) to an external NEMA-3R landscape junction box for future landscape lighting expansion.
- **Under-Awning Future-Proofing:** Pull **14/2 Southwire Duo** to 4 corners of the exterior roofline for future architectural accent lighting.
- **Security Camera Drops:** Ensure all 12 Ubiquiti camera locations (Section 5) have Cat6 home-runs to the Tech Room (LCP-3).

13.4 Closet & Storage Automation (WIC / Linen / Pantry)

- **Reed Switches (Primary):** Install recessed magnetic contact switches (e.g., Tane Pill-V) in all closet door frames. Home-run 18/2 to the nearest LCP Binary Input.
- **Stealth mmWave Pre-Wire:** Pull a KNX Bus "tail" to the center ceiling of each WIC and Linen closet. Coil 2ft of slack **above the drywall** (Stealth Vault).
- **Future sensor:** Prepared for **Creatrol 24G/60G mmWave** hidden installation (drywall penetration mode).
- **Requirement:** NO physical light switches are to be installed in closets or small storage areas. Lighting must be 100% sensor/contact driven.